

Phase 3: Block-Level Design

Marklin WiFi AC Train Controller

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Test Project: Block-Level Design

Block Decomposition

The architecture (Phase 2) defines two galvanically isolated domains - AC (track-referenced) and DC (MCU GND-referenced) - bridged by optocouplers. The design decomposes into 9 schematic blocks, each suitable for a single KiCad hierarchical sheet.

#	Block	Domain	Sheet
1	Power Supply	Crosses boundary (AC in, DC out)	power_supply.kicad_sc
2	ESP32-C3 MCU	DC	h_esp32_c3.kicad_sch
3	Zero-Crossing	Crosses boundary (AC sense, DC	zero_crossing.kicad_sc
4	Detector Hardware Interlock	output) DC	h_hw_interlock.kicad_sch
5	TRIAC Drive (x2)	Crosses boundary (DC control, AC	triac_drive.kicad_sch
6	Voltage Sensing	switch) Crosses boundary (AC sense, DC ADC)	voltage_sense.kicad_sc
7	Status LED	DC	h_status_led.kicad_sch
8	USB Programming	DC	usb_prog.kicad_sch
9	Connectors	AC	connectors.kicad_sch

Block Descriptions

Block 1: Power Supply

Purpose: Convert 24 VAC track voltage to 3.3 VDC for the MCU and digital circuitry.

Function:

- Bridge rectifier converts 24 VAC to ~30-34 VDC peak (unregulated)
- Input filter capacitor smooths rectified DC
- Buck converter steps down to 3.3 VDC regulated output
- Output decoupling capacitors

Interfaces:

- Inputs: AC_L, AC_N (24 VAC from track, via connectors block)
- Outputs: +3V3 (regulated), DC_GND
- Power rails provided: +3V3, DC_GND

Domain: Crosses isolation boundary - AC input side is track-referenced, DC output side establishes the MCU ground domain.

Design notes:

- The bridge rectifier output is the origin of DC_GND - this is the reference point for the entire DC domain
- Buck converter must handle Vin up to ~35 VDC (from 25 VAC peak) with margin

- Output current budget: ESP32-C3 WiFi active ~240 mA + peripherals ~60 mA = ~300 mA typical, design for 500 mA
- EMI filter on input recommended for conducted emissions

Block 2: ESP32-C3 MCU

Purpose: Central controller - runs Modbus TCP server, WiFi stack, motor control logic, and phase-angle timing.

Function:

- ESP32-C3 RISC-V MCU (QFN32 or module)
- Decoupling capacitors on all VDD pins
- RF matching network and PCB antenna (or module with integrated antenna)
- Boot/reset pushbuttons with RC debounce
- 40 MHz crystal (if bare chip, not needed for module)

Interfaces:

- Power: +3V3, DC_GND
- Inputs: ZC_PULSE (zero-crossing interrupt, from block 3), V_SENSE (analog voltage, from block 6), USB_D+/USB_D- (from block 8)
- Outputs: FWD_CMD, REV_CMD (GPIO to hardware interlock, block 4), LED_CTRL (GPIO to status LED, block 7)
- Bidirectional: WiFi (antenna, on-chip)

Domain: DC only.

Design notes:

- If using bare ESP32-C3 chip: need 40 MHz crystal, RF matching, and flash IC
- If using ESP32-C3 module (e.g., ESP32-C3-MINI-1): crystal, flash, and antenna are integrated - simpler BOM, recommended for first revision
- GPIO allocation (preliminary):
 - GPIO0: ZC_PULSE (input, interrupt-capable)
 - GPIO1: V_SENSE (ADC input)
 - GPIO2: FWD_CMD (output)
 - GPIO3: REV_CMD (output)
 - GPIO4: LED_CTRL (output)
 - GPIO18: USB_D- (fixed)
 - GPIO19: USB_D+ (fixed)
 - GPIO8: Boot mode (active low for download, active high for run)
 - GPIO9: Boot button (directly usable as GPIO after boot)

Block 3: Zero-Crossing Detector

Purpose: Sense AC zero-crossings and provide an isolated logic-level pulse to the MCU for phase-angle timing.

Function:

- AC-side resistor divider limits current into optocoupler LED
- Optocoupler (e.g., EL357N or PC817) provides galvanic isolation
- DC-side pull-up resistor and optional Schmitt trigger conditioning
- Output is a pulse at each zero-crossing (~100/120 Hz for 50/60 Hz AC)

Interfaces:

- Inputs: AC_L, AC_N (track AC, from connectors block)
- Outputs: ZC_PULSE (logic-level signal to ESP32 GPIO)
- Power: +3V3 (DC side pull-up only), DC_GND

Domain: Crosses isolation boundary - AC input, DC output.

Design notes:

- The optocoupler LED conducts on each AC half-cycle near the zero crossing
- Series resistor on AC side sized for ~5-10 mA LED current at 24 VAC peak
- Two anti-parallel diodes or a bridge arrangement may be needed to sense both half-cycles
- DC-side output should be fast enough for accurate zero-crossing timing (<100 us jitter)
- CTR (current transfer ratio) of optocoupler affects output rise time

Block 4: Hardware Interlock

Purpose: Enforce mutual exclusion of FWD and REV TRIAC gate signals in hardware, independent of firmware.

Function:

- Two cross-coupled NAND gates (e.g., 74LVC00 or similar)
- If both FWD_CMD and REV_CMD are asserted simultaneously (firmware fault), both outputs are forced inactive
- Normal operation: only one output active at a time
- Optional: dead-time enforcement (brief all-off period during direction change)

Interfaces:

- Inputs: FWD_CMD, REV_CMD (from ESP32 GPIOs)
- Outputs: FWD_GATE, REV_GATE (to TRIAC drive optocouplers, block 5)
- Power: +3V3, DC_GND

Domain: DC only (sits between MCU GPIOs and TRIAC optocoupler DC-side inputs).

Design notes:

- Truth table:

FWD_CM	REV_CM	FWD_GAT	REV_GAT
0	0	0	0
1	0	1	0
0	1	0	1
1	1	0	0

- Uses a single 74LVC00 (quad 2-input NAND) - only 2 gates needed for cross-coupling, with 2 spare
- 3.3V compatible logic family required (LVC, AHC, or similar)
- Active-high convention: output HIGH = TRIAC gate enabled
- Decoupling cap on VCC

Block 5: TRIAC Drive (x2)

Purpose: Isolated gate drive for the two motor winding TRIACs (forward and reverse).

Function:

- DC-side: optocoupler LED driven from interlock output (FWD_GATE or REV_GATE) via current-limiting resistor
- Optocoupler (e.g., MOC3021 zero-cross type or MOC3041 with built-in zero-cross) provides isolation
- AC-side: optocoupler TRIAC output triggers the main power TRIAC gate
- Main TRIAC (e.g., BTA16-600B or similar) switches track AC to motor winding
- Snubber network (RC) across TRIAC for dV/dt protection

Interfaces:

- Inputs: FWD_GATE or REV_GATE (DC side, from interlock block 4), AC_L (track AC, from connectors)
- Outputs: MOTOR_FWD or MOTOR_REV (AC to motor winding, via connectors block 9)
- Power: +3V3, DC_GND (DC side for optocoupler LED drive only)

Domain: Crosses isolation boundary - DC control input, AC power output.

Design notes:

- Two identical channels - can be a single sheet instantiated twice or one sheet with both channels
- MOC3021 (random-phase) recommended over MOC3041 (zero-cross) because phase-angle control requires firing at arbitrary points in the AC cycle, not only at zero crossings
- TRIAC rating: 600V, 4A minimum (motor current typically <2A, but provide margin)
- Gate resistor between MOC3021 output and TRIAC gate (~360 ohm typical)
- Snubber: 100 nF + 100 ohm in series across TRIAC (values to be refined in detailed design)
- Heat sinking may be needed depending on motor current and duty cycle

Block 6: Voltage Sensing

Purpose: Measure track AC voltage magnitude for the MCU (diagnostics, undervoltage detection, speed calibration).

Function:

- Isolated measurement using either:
 - (a) Resistor divider + optocoupler (simple, low accuracy), or
 - (b) Small isolation transformer + rectifier + divider (better accuracy)
- Output scaled to 0-3.3V range for ESP32 ADC input
- Low-pass filter to smooth rectified signal to a DC level proportional to AC RMS

Interfaces:

- Inputs: AC_L, AC_N (track AC, from connectors)
- Outputs: V_SENSE (analog 0-3.3V to ESP32 ADC, block 2)
- Power: +3V3, DC_GND (DC side)

Domain: Crosses isolation boundary - AC sense input, DC analog output.

Design notes:

- Approach (a) is simpler but nonlinear (optocoupler CTR varies). Suitable for relative measurement / threshold detection
- Approach (b) provides better linearity but adds a transformer. Overkill for this application
- Recommended: approach (a) with software calibration
- ADC input protection: series resistor + clamp diodes to 3.3V and GND

- ESP32-C3 ADC is 12-bit but has limited linearity - adequate for voltage monitoring, not precision measurement

Block 7: Status LED

Purpose: Visual indication of controller state (power, WiFi connected, motor running, fault).

Function:

- Single LED driven from MCU GPIO via current-limiting resistor
- GPIO supports PWM for brightness control or blink patterns

Interfaces:

- Inputs: LED_CTRL (from ESP32 GPIO, block 2)
- Power: +3V3, DC_GND

Domain: DC only.

Design notes:

- Standard LED + resistor (e.g., green LED, 1K resistor for ~1.5 mA at 3.3V)
- ESP32 GPIO can source ~12 mA - direct drive is fine, no transistor needed
- Consider using an LED with integrated resistor to reduce BOM
- Firmware defines blink patterns (out of scope for hardware design, but GPIO must support PWM)

Block 8: USB Programming

Purpose: USB-C connector for firmware programming and debug via ESP32-C3 native USB (CDC/JTAG).

Function:

- USB-C receptacle with proper CC pull-down resistors (5.1K to GND) for device mode
- ESD protection TVS diode array on D+/D-
- Direct connection to ESP32-C3 USB pins (GPIO18=D-, GPIO19=D+)

Interfaces:

- Outputs: USB_D+, USB_D- (to ESP32, block 2)
- Power: VBUS (5V from USB host) - not used for powering the board (track-powered), but may be connected for USB detection
- Ground: USB_GND connected to DC_GND

Domain: DC only.

Design notes:

- No USB-UART bridge IC needed - ESP32-C3 has native USB CDC/JTAG
- USB-C CC resistors: 5.1K on each CC pin to GND (device mode, signals USB 2.0 default power)
- ESD protection is essential - USB port is an external interface exposed to user handling
- Consider USB_VBUS detection on a GPIO for auto-boot into download mode (optional, firmware-level feature)
- USB ground must be common with MCU ground (DC_GND)

Block 9: Connectors

Purpose: Physical interface to the track and motor windings.

Function:

- Track input connector: 2-pin for 24 VAC (AC_L, AC_N)
- Motor output connector: 3-pin for motor common + two windings (MOTOR_COM, MOTOR_FWD, MOTOR_REV)
- Motor common is connected to AC_N (one side of track)

Interfaces:

- Track input: AC_L, AC_N (distributed to power supply, zero-crossing detector, voltage sense, TRIAC drive blocks)
- Motor output: MOTOR_FWD (from TRIAC 1), MOTOR_REV (from TRIAC 2), MOTOR_COM (AC_N)

Domain: AC only.

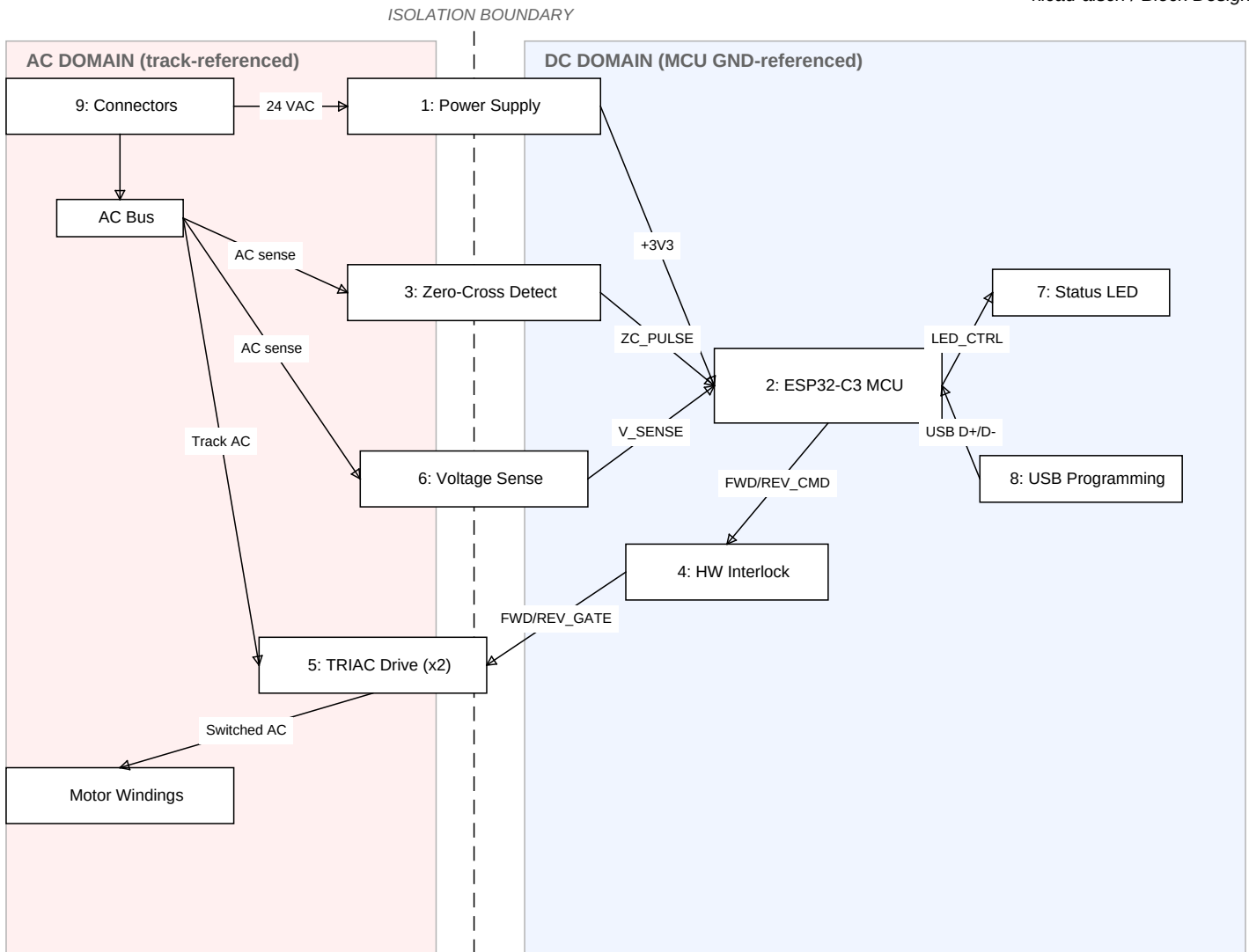
Design notes:

- Connector type depends on mechanical integration (solder pads, pin headers, or wire-to-board)
- Track connections carry up to 2A motor current + MCU supply - wire gauge and connector rating must match
- Motor common (MOTOR_COM) is tied to AC_N - this means one side of both windings shares a common return to the track
- Polarity marking recommended for track input (helps during installation)

Signal and Power Bus Summary

Signal/Rail	Type	From Block	To Block(s)	Domain
AC_L	Power	9 (Connectors)	1, 3, 5, 6	AC
AC_N	Power	9 (Connectors)	1, 3, 5, 6, 9	AC
+3V3	Power	1 (Power Supply)	2, 3, 4, 5, 6, 7, 8	DC
DC_GND	Power	1 (Power Supply)	2, 3, 4, 5, 6, 7, 8	DC
ZC_PULSE	Signal	3 (Zero-Cross)	2 (ESP32 GPIO)	DC
FWD_CMD	Signal	2 (ESP32 GPIO)	4 (Interlock)	DC
REV_CMD	Signal	2 (ESP32 GPIO)	4 (Interlock)	DC
FWD_GATE	Signal	4 (Interlock)	5 (TRIAC Drive)	DC
REV_GATE	Signal	4 (Interlock)	5 (TRIAC Drive)	DC
V_SENSE	Analog	6 (Voltage Sense)	2 (ESP32 ADC)	DC
LED_CTRL	Signal	2 (ESP32 GPIO)	7 (Status LED)	DC
USB_D+/D-	Signal	8 (USB Prog)	2 (ESP32 USB)	DC
MOTOR_FWD	Power	5 (TRIAC 1)	9 (Connectors)	AC
MOTOR_REV	Power	5 (TRIAC 2)	9 (Connectors)	AC

Block Interconnect Diagram



Isolation Boundary Crossings

All signals crossing the AC/DC isolation boundary pass through optocouplers:

Crossing	Direction	Isolation Device	Block
Zero-crossing sense	AC to DC	Signal optocoupler (EL357N/PC817)	3
TRIAC FWD gate drive	DC to AC	TRIAC-output optocoupler (MOC3021)	5
TRIAC REV gate drive	DC to AC	TRIAC-output optocoupler (MOC3021)	5
Voltage sense	AC to DC	Signal optocoupler or isolation amp	6
Power supply	DC to AC	Bridge rectifier + buck (transformer isolation if needed)	1

Total optocouplers: 4 (1x zero-crossing, 2x TRIAC gate drive, 1x voltage sense)

Design Verification Checklist

- All architecture elements from Phase 2 are covered by at least one block
- Each block fits on a single KiCad schematic sheet
- All inter-block interfaces are explicitly named and typed
- Isolation boundary crossings are identified with specific isolation devices
- Power budget accounted for (500 mA at 3.3V from buck converter)

- Hardware interlock truth table ensures mutual exclusion
- Signal flow is unidirectional at each interface (no ambiguous bidirectional signals except USB)
- Motor common return path is clearly defined (AC_N)